

No. of Functional Features Included in the Solution

SMARTBRIDGE - No. of Functional Features Included in the Solution

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-6970
- **Team Size:** 3
- **Team Leader:** Prince Kumar P
- **Team Members:** Gowtham L, Guru Moorthi M
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration	Creates a new user account with encrypted bcrypt password hashing.	Auth Module (authController.js)	Done	0.5 Marks
2	User Login & JWT Auth	Verifies credentials and generates a signed JWT token.	Auth Module (authController.js)	Done	0.5 Marks
3	Get User Profile	Fetches profile details for the authenticated bearer token.	Auth Module (authController.js)	Done	0.5 Marks

4	Add Workout Record	Logs a workout with name, category, duration, calories, and date.	Workout Module (workoutController.js)	Done	0.5 Marks
5	Get / Search Workouts	Retrieves all workouts or filters history by workout name, category, or date.	Workout Module (workoutController.js)	Done	0.5 Marks
6	Update / Delete Workout	Modifies or deletes an existing workout record linked to the user.	Workout Module (workoutController.js)	Done	0.5 Marks
7	AI Workout Plan Generator	Queries Gemini AI (@google/genai) using age, goals, and experience level.	AI Module (aiController.js , geminiService.js)	Done	1.0 Marks
8	AI Fitness Analytics Insights	Analyzes total workouts, avg duration, and calories burned to return insights.	AI Module (aiController.js , geminiService.js)	Done	1.0 Marks

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8 (100%)

Feature Category Breakdown

S.No	Category PDF	Features in Category PDF	Example Features PDF
1	User Interface (UI)	1	REST API JSON Consumer Interface / Postman / Thunder Client
2	Backend / Logic	3	Express Routing, Controller Business Logic, Error Handler Middleware
3	Database / Storage	2	MongoDB Atlas Users Collection, Workouts Collection

4	API / Integration	2	Google Gemini AI Recommendation & Fitness Insight Service Integration
5	Security / Authentication	2	JWT Bearer Authentication, bcryptjs Password Hashing